Shashwat Sourav

Location: St. Louis, MO
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GitHub: Shash12-ship ✻ *Scholar:* Google Scholar

Education

PhD Candidate — *St. Louis, MO, USA*
Washington University in St. Louis — *Aug 2024 - Present*
Physics

B.S.in Data Science — *Bhopal, India*
Indian Institute of Science Education and Research (IISER) Bhopal — *Jan 2021 - July 2024*
GPA: 8.63/10

Positions Held

MARS V Fellow — *July 2026 - Dec 2026*
Cambridge AI Safety Hub — *Oxford, UK (hybrid)*

- Advisors: **Terry Zhang**, Prof. **Zhijing Jin**
- Working on evaluation hacking in AI4Science agentic benchmarks.

GRO Fellow — *Jan 2026 - May 2026*
Oak Ridge National Lab — *Oak Ridge, USA*

- Advisor: **Dr. Tirthankar Ghosal**
- I am working on hypothesis generation and developing AI agents for astronomy and material sciences and developing metrics to validate the generated hypotheses.
- Worked on understanding model hypothesis generation in vision langugage models using circuit tracing.

Bachelor's Thesis Student — *Jan 2024 - April 2024*
Indian Institute of Science — *Bangalore, India*

- Advisors: **Dr. Prathosh AP, Dr. Akshay Agarwal**
- Worked on single-class machine unlearning, focusing on selectively removing the influence of a target class from a pre-trained model. Designed the Partially-Blinded Unlearning (PBU) framework, formulating class unlearning from a Bayesian perspective with Fisher-information–based stability regularization.
- Work accepted at the **39th Association for the Advancement of Artificial Intelligence (AAAI), one of the top conference in AI and ML**.

Research Intern — *Dec 2023 - Aug 2024*
MICA School of Ideas — *Ahmedabad, Gujrat*

- Advisor: **Dr. Praveen SV**
- Worked on natural language processing and sentiment analysis of social-media posts, focusing on Twitter/X discussions of the Gaza conflict in Australia.
- Collected and preprocessed tweets, and analyzed temporal and geographic trends in public opinion across different Australian cities and user communities. Work published in the **International Journal for Disaster Risk Reduction**.

DAAD Working Internships in Science and Engineering Fellow — *May 2023 - Jul 2023*
Universität Heidelberg — *Heidelberg, Germany*

- Advisor: **Prof. Andreas Just**
- I analyzed data from N-body simulations of galaxy mergers in which each galaxy hosts a central supermassive black hole, studying how the merger dynamics shape the black holes' orbits and the surrounding stellar distribution.

Winter Research Fellow *Nov 2022 - Dec 2022*
Arecibo Observatory *Arecibo, Puerto Rico*

- Advisor: **Dr. Selvaraj Dharmalingam**
- Worked on processing 100 datasets based on parameters like wind velocity and meridional velocity to visualize the Gravity Waves generated by Hurricane Maria.

Status: Poster presentation at American Geophysical Union Conference (Chicago, December 2022).

Visiting Summer Research Fellow *May 2022 - August 2022*
National Astronomical Observatory *Mitaka, Japan*

- Advisor: **Prof. Maria Giovanna Dainotti**
- Processed optical afterglow light curves of GRBs, performed broken power-law fits in Mathematica to derive temporal and spectral indices, and tested synchrotron closure relations to identify favored cooling regimes and environments (ISM vs wind), as in recent optical-closure studies.
- Analyzed high-energy GRBs from the Fermi-LAT Second GRB Catalog, extracting temporal and spectral slopes from simple/broken power-law fits and using closure relations in stratified media to distinguish models with and without energy injection.
- Work led to two research papers now published in **The Astrophysical Journal and MDPI Galaxies**.

Research Grants

NASA Nancy Grace Roman Space Telescope Cycle 1 Program *2026*
R-HIVE: Mapping the SFH–Morphology–Environment Connection with Roman

- PI: **Prof. Kartheik Iyer**.
- **Role: Co-Principal Investigator (Co-PI)**.
- Selected as a **Roman Cycle 1 Medium Program** following Science Review Panel and Time Allocation Committee review.

Publications

Do Astronomical Foundation Models Know When They Will Fail? *2026*
V. Bulusu, M. Goyal, R. Sany, M. Smith, ***S.Sourav*** *NeurIPS Interp4Discovery*
Link- `https://openreview.net/pdf?id=kBEhsFs3ox`

Gestalt: a meta-foundation model for astronomy *2026*
M. Smith, ***S.Sourav*** *NeurIPS Interp4Discovery*

Link - `https://arxiv.org/abs/2609.38312v1`

Visualizing Graph-to-Answer Mechanism Recovery in Materials-Science Hypothesis Generation *2026*
S.Sourav*, S.Pal, M.J. Buehler, S. Das, F. Wang, D. Soos, T. Ghosal* *IEEE*

Link - `https://arxiv.org/pdf/2608.04170v1`

Graph-to-Answer Circuits: Mechanistic Interpretability for Materials-Science Hypothesis Generation *2026*
S.Sourav*, S.Pal, M.J. Buehler, T. Ghosal* *ACM AI Summit*

Link - https://arxiv.org/pdf/2607.00924

What AstroPT knows about galaxies, and what that can teach us about LLMs *2026*
K. Duraphe, A. Kumar, M. Smith, ***S.Sourav*** *COLM SciFM*

Link - https://arxiv.org/abs/2608.22614

The Tell-Tale Trace:DetectingReasoning Failures in LLMs Using CoT Dynamics *2026*
S.Sourav*, A. Balwani*

Link - https://arxiv.org/abs/2608.03291

Graph-native Traceable Scientific Hypothesis Generation via Graph-Structured Reinforcement Learning *2026*
S.Pal, ***S.Sourav*** *T. Ghosal, M. Buehler*
Status: **Under Review in Springer Nature Science and AI**

Link - https://arxiv.org/pdf/2607.00924

The Compressive Knowledge Graph Hypothesis: Which Graph Facts Matter for Scientific Hypothesis Generation? *2026*
S.Sourav*, V.Baibakova, S. Das, R.Elgedawy, M.Mahbub, E.Herron, T. Ghosal NeurIPS AI4Mat*

Link - https://arxiv.org/abs/2605.27176

Covariant Energy Density Functionals for Neutron Star Matter Equation of State Modeling: Cross-Comparison Analysis Using CompactObject *2026*
C.Huang, T.Malik, J. Cartaxo, ***S.Sourav****, W.Yuan, T.Zhou, X.Liu, J.Groger, X.Dong, N.Osborn, N.Whitsett, Z.Wang, C.Providência, M.Oertel, AY. Chen, L.Tolos, A.Watts ApJS*

Link - https://iopscience.iop.org/article/10.3847/1538-4365/ae2310

f-INE: A Hypothesis Testing Framework for Estimating Influence under Training Randomness *2026*
S. Panda, D. Tarsadiya, ***S. Sourav****, Prathosh A.P, S.P Karimireddy ICLR*

Link - https://arxiv.org/pdf/2510.10510

The Platonic Universe: Do Foundation Models See the Same Sky? *2025*
K. Duraphe, M. Smith, ***S. Sourav****, John Wu NeurIPS*
Status: **Spotlight Talk (1% papers) at NeurIPS ML4PS workshop**

Link - https://arxiv.org/abs/2509.19453

Cosmology-informed Neural Networks to infer dark energy equation-of-state *2025*
A. Verma, ***S. Sourav****, PK. Aluri, David F. Mota*
Status: **Under Review in JCAP**

Link - https://arxiv.org/abs/2508.12032

A Survey on Hypothesis Generation for Scientific Discovery in the Era of Large Language Models *2025*
Atilla Kaan Alkan, ***Shashwat Sourav*** *, Maja Jablonska, Simone Astarita et. al*

Link - https://arxiv.org/abs/2504.05496

Constraining first-order phase transition inside neutron stars with application of Bayesian techniques on PSR J0437-4715 NICER data *2025*
C. Huang, ***S.Sourav*** *ApJ*

Link - https://iopscience.iop.org/article/10.3847/1538-4357/adbb67

Partially Blinded Unlearning: Class Unlearning for Deep Networks a Bayesian Perspective *2025*
*Subhodip Panda, **Shashwat Sourav**, Prathosh AP* *AAAI*

Paper - https://openreview.net/forum?id=abS0M5Agsi

Analysis of GRB Closure Relationship in Multi-wavelengths *2024*
*M.G Dainotti, S. Bhardwaj, E. Bissaldi, N.Fraija, **S.Sourav**, A. Galvan-Gamez* *ApJ*

Link - https://arxiv.org/abs/2411.10736v

Critique: Sentiment-topic dynamic collaborative analysis-based public opinion mapping in aviation disaster management: A case study of the MU5735 air crash *2024*
*Praveen SV, **Shashwat Sourav**, Ashutosh Dutt* *IJDRR*

Link- https://www.sciencedirect.com/science/article/abs/pii/S221242092400308X

Predicting Missing Light Curves of Gamma-Ray Bursts with Bidirectional-LSTM: An Approach for Enhanced Analysis *2024*
***S.Sourav**, A. Shukla, R.Dwivedi, K.Singh* *SPAICE*

Pre-print - https://zenodo.org/records/13885689

The Closure Relations in Optical Afterglow of Gamma-Ray Bursts *2023*
*M.G Dainotti, D.Levine, N.Fraija, D.Warren, **S.Sourav*** *ApJ*

Link - https://iopscience.iop.org/article/10.3847/1538-4357/ac9b11

The Closure Relations in High-Energy Gamma-ray Bursts Detected by Fermi-LAT *2023*
*M.G Dainotti, D.Levine, N.Fraija, D.Warren, P.Veres, **S.Sourav*** *MDPI Galaxies*

Link- https://www.mdpi.com/2075-4434/11/1/25

Packages Developed

Compact Object: An open-source Python package for full-scope neutron star equation of state inference
*C.Huang, T.Malik, J. Cartaxo, **S.Sourav**, W.Yuan, T.Zhou, X.Liu, J.Groger, X.Dong, N.Osborn, N.Whitsett, Z.Wang, C.Providência, M.Oertel, AY. Chen, L.Tolos, A.Watts*

Github - https://github.com/ChunHuangPhy/CompactObject?tab=readme-ov-file

Technical Reports

CEPC Technical Design Report – Accelerator *2023*
Wallah Abdallah[1], Shashwat Sourav[79], Hongying Zou[284] *Springer Nature*

Report- CEPC

Invited Talks and Poster Presentations

Center For Decoding the Universe Journal Club *Palo Alto,CA*
Stanford University *2025*

- Gave a talk on my paper: **The Platonic Universe: Do Foundation Models See the Same Sky?**
- Slides: *Platonic Universe*

Analysis of GRB Closure Relationships in Multi-wavelengths *MARAC 2024*
*Maria, Dainotti, **S.Sourav**, Shubham Bhardwaj, Elisabetta Bissaldi, Nissim Fraija, Antonio Galvan Gamez*

Poster- GRB Multi Wavelength

Department of Astronomy and Astrophysics *Mumbai,India*
Tata Institute for Fundamental Research *2023*

- Gave a talk on: **The Influence of Source Parameters on Gravitational Waveform Emission from Compact Binaries**
- Slides: Gravitational Waves

The Impact of Hurricane-Maria Generated Gravity Waves on Structure and Dynamics of the UTLS *AGU 2022*
S. Dharmalingam, Christiano Garnett Marques Brum and ***S.Sourav***

Poster- Hurricane Maria

Awards and Honors

Mentorship for Alignment Researchers (MARS) Fellow *April 2026*
Selected among the top 75 candidates from over 1000 applications for the MARS fellowship offered by Cambridge AI Safety Hub.

LSST-DA Data Science Fellowship *April 2026*
Selected for a fellowship focused on astronomy data science for the Rubin/LSST era.

Graduate Research Opportunities (GRO) Fellowship *Jan 2026*
Prestigious Fellowship to work at Oak Ridge National Lab in spring semester 2026.

WashU Arts and Sciences Graduate Research Fellowship *Aug 2024*
Graduate Research Assistant fellowship at WashU

DAAD WISE Fellowship *Feb 2023*
Funded research internship at Universitat Heidelberg

NAOJ Travel Grant *April 2022*
Fully funded research internship at National Astronomical Observatory of Japan *Mitaka, Japan*

Gold Honor: International Astronomy and Astrophysics Competition *June 2021*
Secured a place among the Top 1% student selected for the Final Round.

National Initiative for Undergraduate Sciences Fellowship *April 2021*
Fully funded research internship at ARIES *Nainital,India.*

Technical skills

Programming Languages	C++, Python, MATLAB
Data Handling	X-ray data- Chandra Observatory; UV data - FOC (Faint Object Camera); Hubble Space Telescope, Fermi-LAT data, CTA data.
Statistical Techniques	Bayesian Statistics.
Coding on Python	Numerical Simulations, Machine Learning, Deep Learning.
Other Tools	CLOUDY, Mathematica, LATEX.
Frameworks	GIT, Windows, Fedora, VS Code, Spyder
Libraries	ScikitLearn, Pytorch, Keras, Tensorflow, Gammapy, ctools

Mentoring

Research Mentor – Summer of Open AI Research (SOAR) *Jul 2026 – Aug 2026*
EleutherAI *Remote*

- Mentees: **Vaidehi Bulusu**, **Mehul Goyal**, and **Rabius Sany**.
- Research project: **Do Astronomical Foundation Models Know When They Will Fail?**

Hackathons

Amazon ML Challenge Finalist *April 2023*
Amazon *India*

- Participated in the global ML challenge competition with other 10,000 teams organized by Amazon.

Pravega Hackathon Runner's Up *Dec 2021*
Indian Institute of Science, Bangalore *Bangalore,India*

- Runners-Up among the top 10 teams selected for the final round from 10,000 students that participated in the hackathon.

Teaching Experience

Assistant to the Instructor (AI) *Fall 2025*
WashU *St. Louis*

- **Course: Introduction to Astrophysics (PHY 206)**
- Worked as an AI with Dr. Yajie Yuan.
- Taught a batch of 30 second year undergraduate students the basics of astronomy, starting from radiative transfer to black holes and high energy astrophysical phenomena such as gamma-ray bursts.

Assistant to the Instructor (AI) *Spring 2025*
WashU *St. Louis*

- **Course: Introductory Physics Laboratory (PHY 192)**
- Worked as an AI with Dr. Merita Haxhia.
- Taught a batch of 50 first year undergraduate students different physics experiments such as calculating electric field and potential, simple electrical circuits .

Teaching Assistant *Spring 2023*
IISER Bhopal *India*

- **Course: Digital Image Processing (DSE 409/609)**
- Worked as a Teaching Assistant under Prof. Bhavna Rajasekaran.
- Taught a class of 100 senior undergraduates and graduate students about algorithms like SIFT and SIFT-RANSAC and Convolutional Neural Networks (CNNs).
- Organized weekly programming sessions and quizzes to encourage critical thinking and help students in developing logic to write codes in python.

Teaching Assistant *Fall 2023*
IISER Bhopal *India*

- **Course: Data Science in Practice (DSE 315/615)**
- Worked as a Teaching Assistant under Prof. Bhavna Rajasekaran.
- Taught Python fundamentals and how to implement supervised Machine Learning Algorithms.

- Worked on taking hands-on programming sessions and designed questions for assignments.

Reviewer

Journal: The Astrophysical Journal(ApJ)

Conference: 41st Association for the Advancement of Artificial Intelligence (AAAI) 2027
40th Association for the Advancement of Artificial Intelligence (AAAI) 2026
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS) 2025

Workshop: The Fourthieth Annual Conference on Neural Information Processing Systems (NeurIPS) AI4Good Workshop 2026
The Fourthieth Annual Conference on Neural Information Processing Systems (NeurIPS) Interp4Discovery Workshop 2026